

I3 Numerical integration A level only

Name: _____

Class: _____

Date: _____

Time: **226 minutes**

Marks: **189 marks**

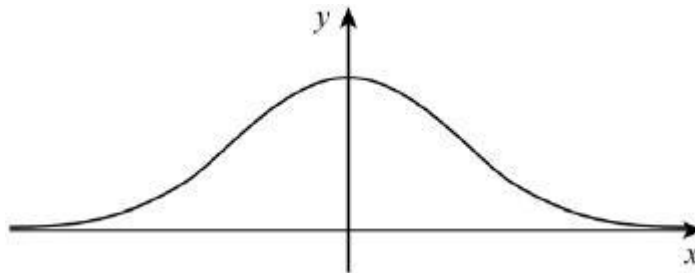
Comments:

EXAM PAPERS PRACTICE

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Q1.

The diagram shows part of the graph of $y = e^{-x^2}$

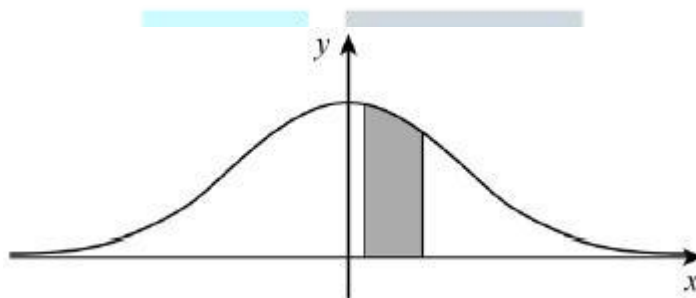


The graph is formed from two convex sections, where the gradient is increasing, and one concave section, where the gradient is decreasing.

- (a) Find the values of x for which the graph is concave.

(4)

- (b) The finite region bounded by the x -axis and the lines $x = 0.1$ and $x = 0.5$ is shaded.



Use the trapezium rule, with 4 strips, to find an estimate for $\int_{0.1}^{0.5} e^{-x^2} dx$
 Give your estimate to four decimal places.

(3)

- (c) Explain with reference to your answer in part (a), why the answer you found in part (b) is an underestimate.

(2)

- (d) By considering the area of a rectangle, and using your answer to part (b), prove that the shaded area is 0.4 correct to 1 decimal place.

(3)

(Total 12 marks)

Q2.

- (a) Use the trapezium rule with five ordinates (four strips) to find an approximate value for

$$\int_1^5 \frac{1}{x^2 + 1} dx$$

giving your answer to three significant figures.

(4)

- (b) (i) Find $\int \left(x^{-\frac{3}{2}} + 6x^{\frac{1}{2}} \right) dx$, giving the coefficient of each term in its simplest form.

(3)

- (ii) Hence find the value of $\int_1^4 \left(x^{-\frac{3}{2}} + 6x^{\frac{1}{2}} \right) dx$.

(2)

(Total 9 marks)

Q3.

- (a) Use the trapezium rule with five ordinates (four strips) to find an approximate value for $\int_0^2 \sqrt{8x^3 + 1} dx$, giving your answer to three significant figures.

(4)

- (b) Describe the single transformation that maps the graph of $y = \sqrt{8x^3 + 1}$ onto the graph of $y = \sqrt{x^3 + 1}$.

(2)

- (c) The curve with equation $y = \sqrt{x^3 + 1}$ is translated by $\begin{bmatrix} 2 \\ -0.7 \end{bmatrix}$ to give the curve with equation $y = g(x)$. Find the value of $g(4)$.

(3)

(Total 9 marks)

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Q4.

- (a) Use the trapezium rule with five ordinates (four strips) to find an approximate value for

$$\int_0^4 \frac{2^x}{x+1} dx$$

giving your answer to three significant figures.

(4)

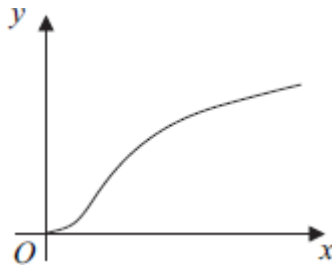
- (b) State how you could obtain a better approximation to the value of the integral using the trapezium rule.

(1)

(Total 5 marks)

Q5.

The diagram shows part of a curve whose equation is $y = \log_{10}(x^2 + 1)$.



- (a) Use the trapezium rule with five ordinates (four strips) to find an approximate value for

$$\int_0^1 (\log_{10}(x^2 + 1)) \, dx$$

giving your answer to three significant figures.

(4)

- (b) The graph of $y = 2 \log_{10} x$ can be transformed into the graph of $y = 1 + 2 \log_{10} x$ by means of a translation. Write down the vector of the translation.

(1)

- (c) (i) Show that $\log_{10}(10x^2) = 1 + 2 \log_{10} x$.

(2)

- (ii) Show that the graph of $y = 2 \log_{10} x$ can also be transformed into the graph of $y = 1 + 2 \log_{10} x$ by means of a **stretch**, and describe the stretch.

(4)

- (iii) The curve with equation $y = 1 + 2 \log_{10} x$ intersects the curve $y = \log_{10}(x^2 + 1)$ at the point P . Given that the x -coordinate of P is positive, find the gradient of the line OP , where O is the origin. Give your answer in the form $\log_{10} \left(\frac{a}{b} \right)$, where a and b are integers.

(4)

(Total 15 marks)

Q6.

- (a) Use the trapezium rule with four ordinates (three strips) to find an approximate value for $\int_0^{1.5} \sqrt{27x^3 + 4} \, dx$, giving your answer to three significant figures.

(4)

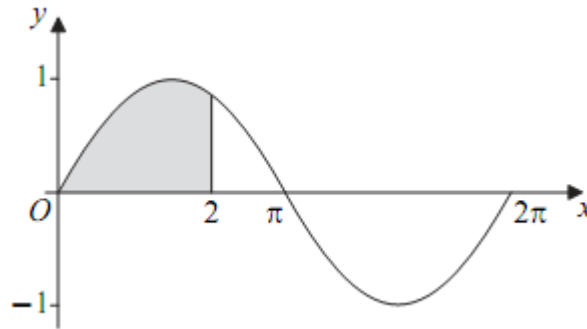
- (b) The curve with equation $y = \sqrt{27x^3 + 4}$ is stretched parallel to the x -axis with scale factor 3 to give the curve with equation $y = g(x)$. Write down an expression for $g(x)$.

(2)

(Total 6 marks)

Q7.

A curve C , defined for $0 \leq x \leq 2\pi$ by the equation $y = \sin x$, where x is in radians, is sketched below. The region bounded by the curve C , the x -axis from 0 to 2 and the line $x = 2$ is shaded.



- (a) The area of the shaded region is given by $\int_0^2 \sin x \, dx$, where x is in radians.

Use the trapezium rule with five ordinates (four strips) to find an approximate value for the area of the shaded region, giving your answer to three significant figures.

(4)

- (b) Describe the geometrical transformation that maps the graph of $y = \sin x$ onto the graph of $y = 2 \sin x$.

(2)

- (c) Use a trigonometrical identity to solve the equation

$$2 \sin x = \cos x$$

in the interval $0 \leq x \leq 2\pi$, giving your solutions in radians to three significant figures.

(4)

(Total 10 marks)

Q8.

- (a) Sketch the curve with equation $y = 2^x$, indicating the coordinates of any point where the curve intersects the coordinate axes.

(2)

- (b) (i) Use the trapezium rule with five ordinates (four strips) to find an approximate value for $\int_0^2 2^x \, dx$, giving your answer to three significant figures.

(4)

- (ii) State how you could obtain a better approximation to the value of the integral using the trapezium rule.

(1)

- (c) Describe a geometrical transformation that maps the graph of $y = 2^x$ onto the graph of $y = 2^{x+7} + 3$.

(3)

- (d) The curve $y = 2^{x+k} + 3$ intersects the y -axis at the point $A(0, 8)$.

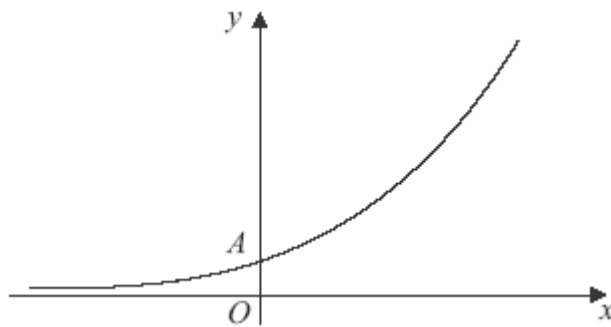
Show that $k = \log_m n$, where m and n are integers.

(2)

(Total 12 marks)

Q9.

The diagram shows a sketch of the curve $y = 2^{4x}$.



The curve intersects the y -axis at the point A .

- (a) Find the value of the y -coordinate of A .

(1)

- (b) Use the trapezium rule with six ordinates (five strips) to find an approximate value for $\int_0^1 2^{4x} dx$, giving your answer to two decimal places.

(4)

- (c) Describe the geometrical transformation that maps the graph of $y = 2^{4x}$ onto the graph of $y = 2^{4x-3}$.

(2)

- (d) The curve $y = 2^{4x}$ is translated by the vector $\begin{bmatrix} 1 \\ -\frac{1}{2} \end{bmatrix}$ to give the curve $y = g(x)$.

The curve $y = g(x)$ crosses the x -axis at the point Q . Find the x -coordinate of Q .

(4)

- (e) (i) Given that

$$\log_a k = 3 \log_a 2 + \log_a 5 - \log_a 4$$

show that $k = 10$.

(3)

- (ii) The line $y = \frac{5}{4}$ crosses the curve $y = 2^{4x-3}$ at the point P . Show that the

x -coordinate of P is $\frac{1}{4 \log_{10} 2}$.

(3)

(Total 17 marks)

Q10.

- (a) Use the trapezium rule with four ordinates (three strips) to find an approximate value for

$$\int_{1.5}^6 x^2 \sqrt{x^2 - 1} \, dx$$

giving your answer to three significant figures.

(4)

- (b) State how you could obtain a better approximation to the value of the integral using the trapezium rule.

(1)

(Total 5 marks)

Q11.

- (a) Use the trapezium rule with four ordinates (three strips) to find an approximate value for $\int_0^6 \sqrt{x^3 + 1} \, dx$, giving your answer to four significant figures.

(4)

- (b) The curve with equation $y = \sqrt{x^3 + 1}$ is stretched parallel to the x -axis with scale

factor $\frac{1}{2}$ to give the curve with equation $y = f(x)$. Write down an expression for $f(x)$.

(2)

(Total 6 marks)

Q12.

Use the trapezium rule with four ordinates (three strips) to find an approximate value for

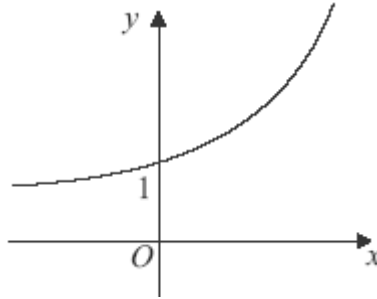
$$\int_0^3 \sqrt{x^2 + 3} \, dx$$

giving your answer to three decimal places.

(Total 4 marks)

Q13.

The diagram shows a sketch of the curve with equation $y = 6^x$.



- (a) (i) Use the trapezium rule with five ordinates (four strips) to find an approximate value for $\int_0^2 6^x dx$, giving your answer to three significant figures. (4)
- (ii) Explain, with the aid of a diagram, whether your approximate value will be an overestimate or an underestimate of the true value of $\int_0^2 6^x dx$. (2)
- (b) (i) Describe a single geometrical transformation that maps the graph of $y = 6^x$ onto the graph of $y = 6^{3x}$. (2)
- (ii) The line $y = 84$ intersects the curve $y = 6^{3x}$ at the point A . By using logarithms, find the x -coordinate of A , giving your answer to three decimal places. (4)
- (c) The graph of $y = 6^x$ is translated by $\begin{bmatrix} 1 \\ -2 \end{bmatrix}$ to give the graph of the curve with equation $y = f(x)$. Write down an expression for $f(x)$. (2)

(Total 14 marks)

Q14.

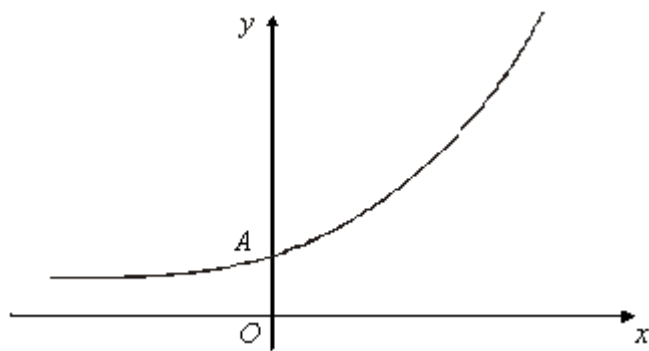
Use the trapezium rule with four ordinates (three strips) to find an approximate value for

$\int_0^3 \sqrt{2^x} dx$ giving your answer to three decimal places.

(Total 4 marks)

Q15.

The diagram shows a sketch of the curve with equation $y = 3(2^x + 1)$.



The curve $y = 3(2^x + 1)$ intersects the y -axis at the point A .

- (a) Find the y -coordinate of the point A . (2)
- (b) Use the trapezium rule with four ordinates (three strips) to find an approximate value for $\int_0^6 3(2^x + 1) dx$ (4)
- (c) The line $y = 21$ intersects the curve $y = 3(2^x + 1)$ at the point P .
 - (i) Show that the x -coordinate of P satisfies the equation $2^x = 6$ (1)
 - (ii) Use logarithms to find the x -coordinate of P , giving your answer to three significant figures. (3)

(Total 10 marks)

Q16.

- (a) Use the trapezium rule with five ordinates (four strips) to find an approximate value for

$$\int_0^4 \frac{1}{x^2 + 1} dx$$

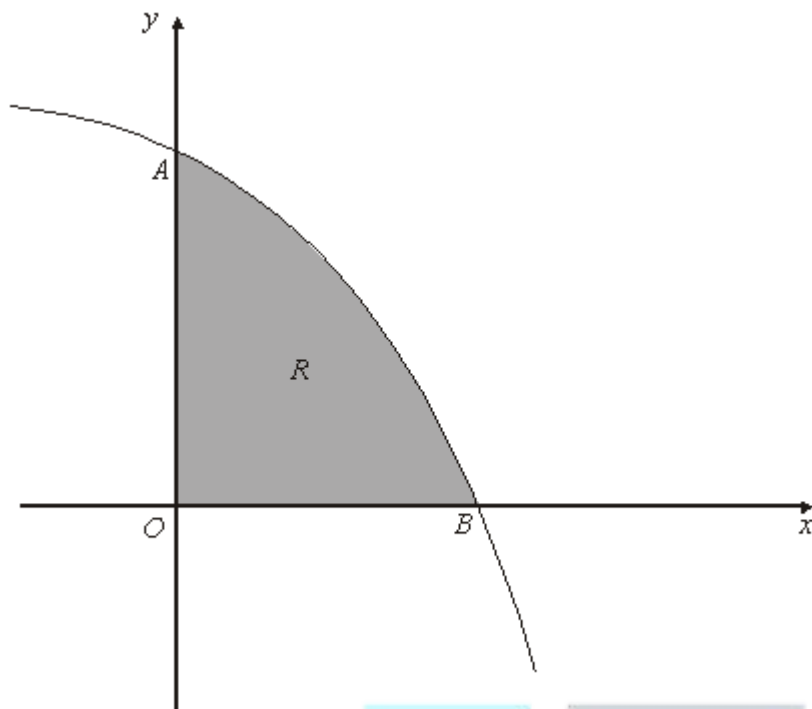
giving your answer to four significant figures.

- (b) State how you could obtain a better approximation to the value of the integral using the trapezium rule.

(1)
(Total 5 marks)

Q17.

The diagram shows a sketch of the curve with equation $y = 27 - 3^x$.



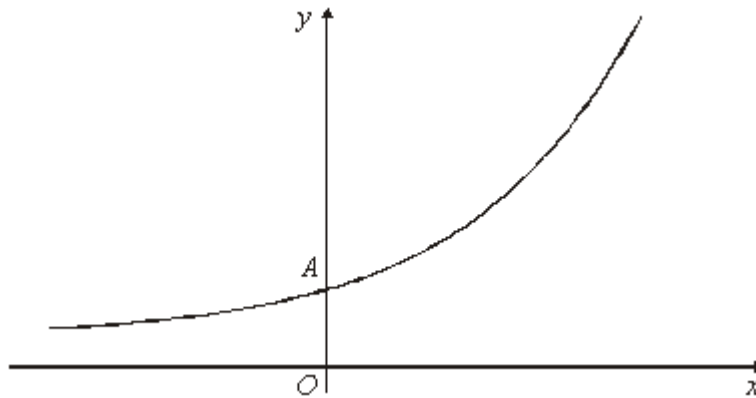
The curve $y = 27 - 3^x$ intersects the y -axis at the point A and the x -axis at the point B .

- (a) (i) Find the y -coordinate of point A . (2)
- (ii) Verify that the x -coordinate of point B is 3. (1)
- (b) The region, R , bounded by the curve $y = 27 - 3^x$ and the coordinate axes is shaded. Use the trapezium rule with four ordinates (three strips) to find an approximate value for the area of R . (4)
- (c) (i) Use logarithms to solve the equation $3^x = 13$, giving your answer to four decimal places. (3)
- (ii) The line $y = k$ intersects the curve $y = 27 - 3^x$ at the point where $3^x = 13$.
Find the value of k . (1)
- (d) (i) Describe the single geometrical transformation by which the curve with equation $y = -3^x$ can be obtained **from** the curve $y = 27 - 3^x$. (2)
- (ii) Sketch the curve $y = -3^x$. (2)

(Total 15 marks)

Q18.

The diagram shows a sketch of the curve with equation $y = 3^x + 1$.



The curve intersects the y -axis at the point A .

- (a) Write down the y -coordinate of point A . (1)

- (b) (i) Use the trapezium rule with five ordinates (four strips) to find an approximation for $\int_0^1 (3^x + 1) dx$, giving your answer to three significant figures. (4)

- (ii) By considering the graph of $y = 3^x + 1$, explain with the aid of a diagram whether your approximation will be an overestimate or an underestimate of the true value of $\int_0^1 (3^x + 1) dx$. (2)

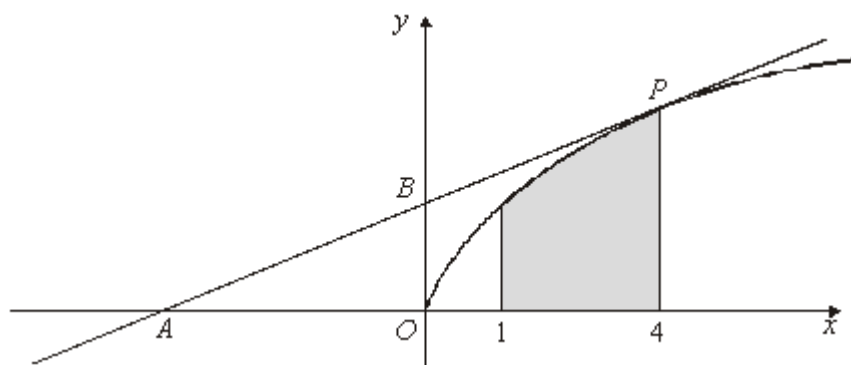
- (c) The line $y = 5$ intersects the curve $y = 3^x + 1$ at the point P . By solving a suitable equation, find the x -coordinate of the point P . Give your answer to four decimal places. (4)

- (d) The curve $y = 3^x + 1$ is reflected in the y -axis to give the curve with equation $y = f(x)$. Write down an expression for $f(x)$. (1)

(Total 12 marks)

Q19.

The diagram shows a curve C with equation $y = \sqrt{x}$. The point O is the origin $(0, 0)$.



The region bounded by the curve C , the x -axis and the vertical lines $x = 1$ and $x = 4$ is shown shaded in the diagram.

- (a) (i) Write $\sqrt{x}$ in the form x^p , where p is a constant. (1)
- (ii) Find $\int \sqrt{x} \, dx$. (2)
- (iii) Hence find the area of the shaded region. (3)
- (b) The point on C for which $x = 4$ is P . The tangent to C at the point P intersects the x -axis and the y -axis at the points A and B respectively.
- (i) Find an equation for the tangent to the curve C at the point P . (4)
- (ii) Find the area of the triangle AOB . (3)
- (c) Describe the single geometrical transformation by which the curve with equation $y = \sqrt{x-1}$ can be obtained from the curve C . (2)
- (d) Use the trapezium rule with four ordinates (three strips) to find an approximation for $\int_1^4 \sqrt{x-1} \, dx$, giving your answer to three significant figures. (4)

(Total 19 marks)